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SARDAR PATEL UNIVERSITY
M.Sc.(CA&IT) Examination, 3rd Semester
Friday, 1st January, 2021.
Time: 10:00 A.M to 12:00 P.M
Subject Code: PS03EHT32
Subject: Parallel Computing

Total Marks: 70**Q.1 Multiple Choice Questions****[10]**

- [1] The Maximum distance between two processors in the computer system is.
A. Bandwidth. C. Bisectionwidth.
B. diameter. D. Network size.
- [2] The number of task that can be executed in parallel is
A. Concurrency C. Degree
B. Degree of Concurrency D. Parallel Programming
- [3] The _____ of a device is the number of result it produce per unit time.
A. Pipeline. B. Segments. C. Throughput. D. Speedup.
- [4] _____ are determine by taskdependency and task interaction graph.
A. Processing C. Both A and B
B. Mapping D. None of Above
- [5] _____ Task are generated as we perform computation.
A. Static C. Simple
B. Dynamic D. Both A and B
- [6] Flits means
A. Flow Control System C. Flow Control Digit
B. Flow Cut digit D. Flow Control Device
- [7] The dual of one to all broadcast is
A. All-all Reduction C. All-one reduction
B. One-all reduction D. One-one reduction
- [8] Single node send a unique meseage of size m to every other node.
A. Gather C. One-all communication
B. Scatter D. All-all Broadcast
- [9] The Maximum distance between two processors in the computer system is
A. Bandwidth C. Networksize
B. Diameter D. Bisectionwidth
- [10] MISD means
A. Multiprocessor instruction single data C. Multiple Instruction Single Data
B. Mutiple Instruction System Data D. Multi Instruction Single Data

Q.2 Fill in the blanks or True/False**[08]**

1. The Throghput of a device is the number of result it produce per unit time.(True/False)
2. NUMA means _____
3. EREW means _____
4. Static Network are referred as indirect network(True/False)
5. Per hope time means time is a function of number of hops and includes factors such as switch latencies, network delays, etc.(True/False)
6. In _____ dependency the result of one operation is an input to the next.
7. Time spent at sending and receiving nodes is known as per- hop time.(True/False)
8. Dynamic networks are also referred to as _____ networks.

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[PTA.3]

Q.3 Short Questions(Attempt Any Ten)

[20]

1. What is task dependency graph?
2. What is critical path length?
3. List decomposition technique.
4. List down mapping techniques.
5. What is Parallel Processing.
6. Define Term.Super computer and Degree of Parallelism
7. Draw the Diagram for Taxonomy of Parallel Processor Architecture.
8. What is Startup time and Per-hop time.
9. What do you mean by all-all broadcast.
10. What is scatter?
11. List characteristics of graph.
12. Define diameter.

Q.4 Long Questions(Attempt Any Four)

[32]

1. List all Topology. Explain any three in detail.
2. Explain Parallelism in Uniprocessor system.
3. Explain Superscalar Execution.
4. Explain architecture of Ideal Parallel Computer(PRAM).
5. Explain characteristics of graph.
6. List and Explain static Mapping Scheme.
7. Explain one to all broadcast and all to one reduction in detail.
8. Write a note on Prefix Sum Operation.

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